

Technical Design & Architecture Document

SALESFORCE DATA CLOUD SEMANTIC MODELING & AGENTFORCE IMPLEMENTATION SPECIFICATION

1. Executive Summary & Data Ingestion Pipeline

This technical document explains how customer master records and web activity data (clickstream) are imported, processed, and utilized. The system maps different tables inside Salesforce Data Cloud, calculates a scoring system for user engagement, and passes this data to an autonomous Agentforce Advisory Agent to provide next-best-action recommendations.

The data layer automatically imports customer profiles from two separate sources directly into Data Lake Objects (DLOs) inside Salesforce Data Cloud:

- **Salesforce CRM Customer Data Stream:** Imports master registration profiles containing basic customer details, account references, phone numbers, and subscription markers.
- **Web Engagement Data Stream (Myntra Storefront):** Imports real-time storefront activity data, including page views, clicks, items added to wishlists, shopping carts, and user login status (Known vs. Unknown users).

2. Standard Data Modeling & Identity Clean-up

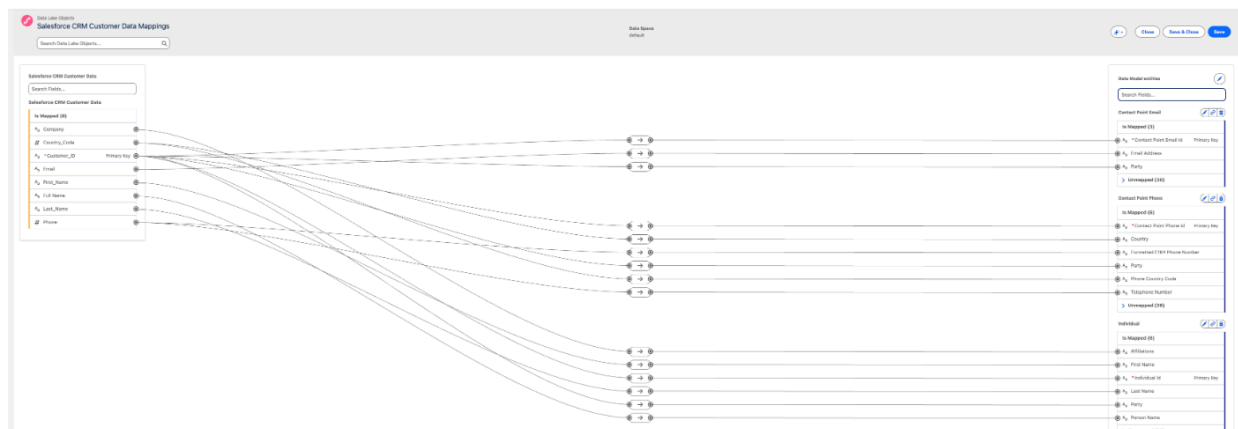
To combine web activity with CRM master records into a single unified identity, data from the DLOs is mapped using the Data Cloud Mapping Engine into standard Data Model Objects (DMOs):

- **Individual DMO:** Serves as the main anchor for personal profiles, linking web sessions from different devices to a single master customer record.
- **Contact Point Email DMO:** Organizes email addresses and matches records to find and remove duplicate profiles.
- **Contact Point Phone DMO:** Organizes phone numbers and cleans up raw number formatting across data sources.

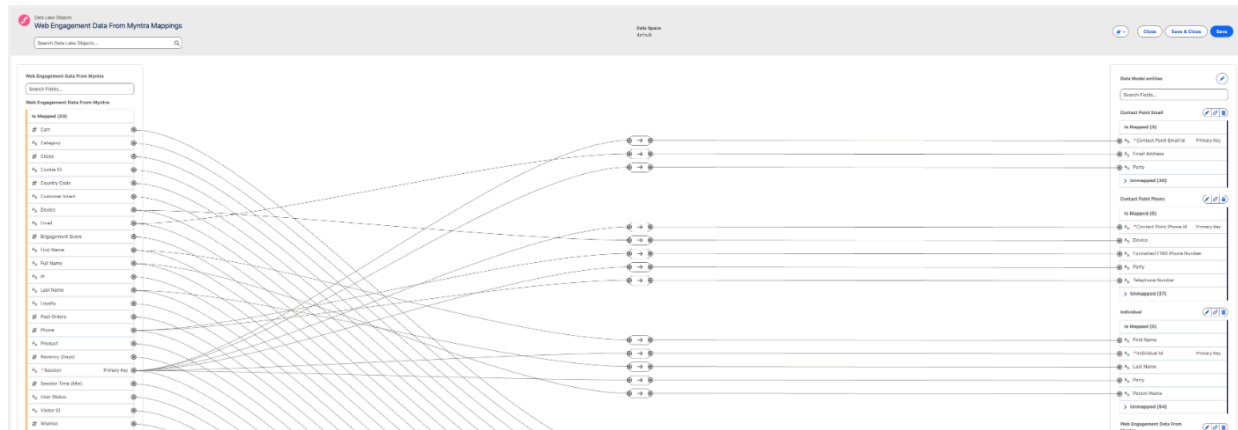
Official Mapping Guide Reference

Official Mapping Guide Followed: https://help.salesforce.com/s/articleView?id=data.c360_a_required_data_mappings.htm&type=5

Salesforce CRM Customer Data Mappings



Web Engagement Data From Myntra Mappings



3. Calculated Insights Configuration

To deliver immediate analytics to the frontend Lightning Web Component (LWC), three distinct Calculated Insights (CI) modules were built using SQL queries to group and calculate data:

- **Calculated Insight 1 (Traffic Segmentation Split):** Calculated Insight 1 (Traffic Segmentation Split): Calculates the average engagement score and total count of active unique sessions, split by user state (Known vs. Unknown). This highlights the activity differences between anonymous browsers and logged-in shoppers.
- **Calculated Insight 2 (Unified Individual Engagement):** Groups and consolidates multi-device web activity back to a single unified customer profile. It adds up total cart additions, wishlists, and order histories to show cross-channel customer health.
- **Calculated Insight 3 (Customer Intent Category Persona Matrix):** Evaluates real-time shopping behavior and cart updates to automatically assign customers into specific marketing intent groups.

4. Algorithmic Processing & Performance Reviews

A. System Data Normalization Bounds

To keep the calculations reliable and lock output scores safely within a standard 0 to 100 scale, incoming data points are restricted to these operational ranges:

- **Storefront Clicks:** Restricted to a standard range of 0 to 100 clicks.
- **Wishlist Additions:** Bounded between 0 to 10 items.
- **Active Shopping Cart Items:** Restricted to a range of 0 to 10 items in the cart.
- **Historical CRM Purchases:** Bounded between 0 to 10 orders.
- **Session Browsing Times:** Normalized between 0 to 60 minutes.
- **Interaction Recency:** Freshness windows tracked across a 0 to 30 days timeline.

B. Engagement Score Proportional Weighting Matrix

	A	B	C	D
1	Storefront Metric Track	Max Cap	Weight	Core Strategic Weightage Rationale
2	Cart Additions (Cart)	10 Items	25%	Highest Intent: Adding items to an active checkout cart is the strongest behavioral signal that a user is ready to buy.
3	Past Purchase Orders (Orders)	10 Orders	20%	Loyalty Factor: Past order fulfillment proves brand trust; returning buyers demonstrate a higher conversion baseline.
4	Wishlist Additions (Wishlist)	10 Items	15%	Future Intent: Shows specific interest in targeted products, serving as a great anchor for personalized offers.
5	Session Browsing Duration (Time)	60 Mins	15%	Brand Stickiness: Measures focused attention span and site exploration depth over simple quick clicks.
6	Interaction Recency Freshness (Recency)	30 Days	15%	Freshness Decay: Activity logged within the last 24 hours marks a hot lead; the score decreases linearly as time passes.
7	Click Volume Frequency (Clicks)	100 Clicks	10%	General Activity: Tracks baseline site visibility but can indicate navigation confusion if decoupled from cart additions.

C. Engagement Score Algorithmic Modeling Formula

Standardized Engagement Score Formula

EngagementScore = 25 * (Cart / 10) + 20 * (Orders / 10) + 15 * (Wishlist / 10) + 15 * (Time / 60) + 15 * (1 - Recency / 30) + 10 * (Clicks / 100)

D. Customer Intent Core Calculation Modeling

To turn numbers into clear service advice, the system calculates customer intent categories by passing the engagement score and CRM loyalty tier data into a nested conditional statement:

Logical Customer Intent Selection Formula

```
=IF(EngagementScore >= 70, "High Intent: Purchase Ready",  
IF(AND(EngagementScore >= 40, Loyalty = ""), "High Intent: New Lead",  
IF(AND(EngagementScore >= 40, Loyalty <> ""), "Mid Intent: Retention",  
IF(EngagementScore >= 25, "Low Intent: Researcher", "Passive: Window Shopper"))))
```

Strategic Segmentation Rationale & Logic Breakdown:

- **High Intent (Purchase Ready):** Any customer scoring 70 or higher is marked as an immediate outreach priority, completely bypassing loyalty filters. At this level, active shopping carts and browsing patterns prove they are ready to buy right now.
- **High Intent (New Lead):** Triggered when a high engagement score (>= 40) combines with an empty CRM Loyalty field (Loyalty = ""). This highlights a high-value anonymous user exploring the store—marking the perfect time to offer a first-time account setup discount or capture an email.
- **Mid Intent (Retention):** Runs when a strong engagement score (>= 40) is paired with an active, populated loyalty tier (Loyalty <> "", like Gold or Platinum). This identifies returning fans who respond best to loyalty rewards or exclusive product access.
- **Low Intent / Researcher / Window Shopper:** Marked by lower scoring tiers (25 to 39 for Researchers, and < 25 for passive Window Shoppers). This indicates generalized browsing or site exploration without any clear purchase intent.

5. Agentforce Advisory Configuration & Grounding

The top orchestration layer configures an autonomous Agentforce Advisory Agent equipped with subagents specialized in real-time customer context evaluation. When an employee searches the customer name inside preview, the subagent analyzes current tracking parameters and triggers the grounded recommendation action sequence.

This subagent calls the custom Einstein Prompt Template Advisory_Prompt, passing the string variable Input:Customer_Name to anchor generation parameters. The template uses fine-tuning prompt guidelines to instruct the Einstein LLM to query the registered vector retriever, look up the consolidated customer footprint data, parse current Calculated Insights matrices, and isolate a targeted, next-best-action text string block without generic background boilerplate padding.

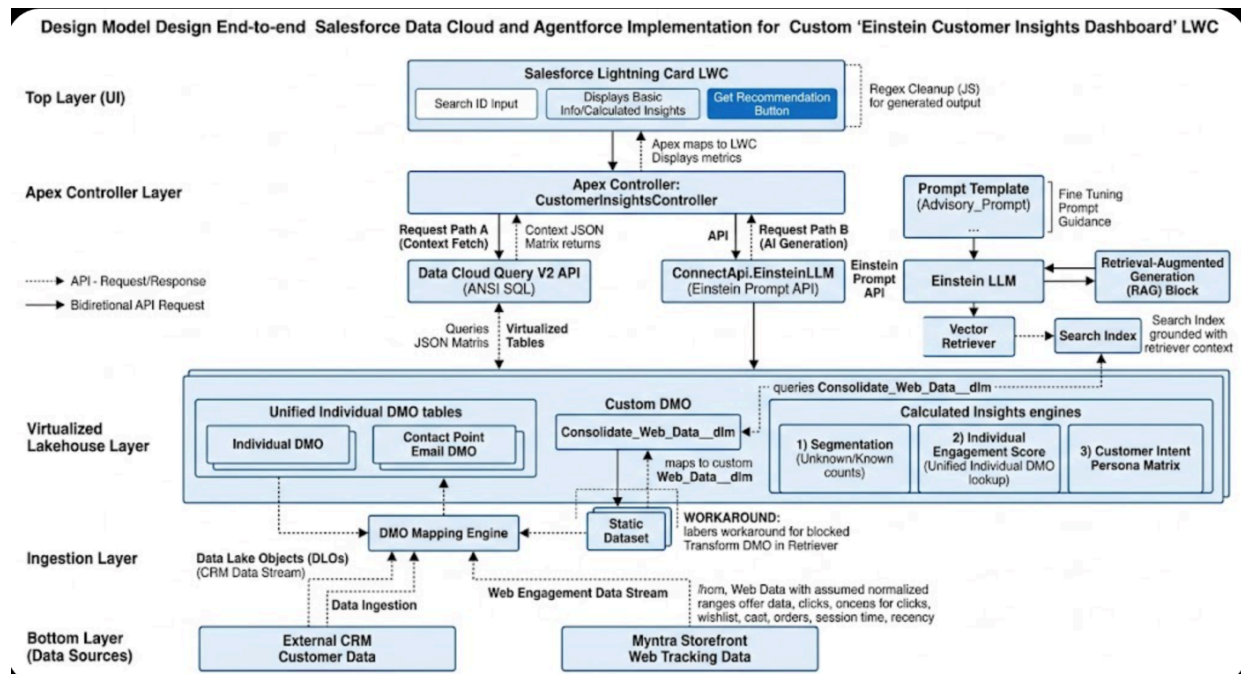
6. Displaying Data on the LWC UI

To show the generated recommendations on the custom user interface, the system runs an asynchronous, imperative call from the Lightning Web Component to the custom Apex controller, which connects to the ConnectApi.EinsteinLLM programmatic gateway. This backend broker wraps inputs inside ConnectApi.WrappedValue to prevent prompt injection before calling the prompt template.

Once the nested ConnectApi.EinsteinPromptTemplateGenerationsRepresentation text payload is successfully returned to the LWC framework, a client-side JavaScript regular expression seamlessly processes the raw string buffer to strip away background reasoning boilerplate text. This sanitized recommendation is then dynamically injected into the browser DOM using conditional rendering rules inside a high-contrast UI box, allowing users to review real-time advice instantly without hurting page performance.

7. End-to-End System Architecture Blueprint

The technical layout traces ingested records from external data channels down to the canonical DMO schemas, custom workaround landing spaces, indexing retrievers, and dual transaction execution channels managed by the Apex controller layer. (Please refer to the technical design model layout blueprint for visual tracing references).



Platform Engineering Alert: Data Transform Runtime Retrieval Defect

Symptom: Data Model Objects (DMOs) created as outputs from standard Data Cloud Streaming or Batch Data Transforms can be successfully Search Indexed, chunked, and mapped to a Vector Retriever without any errors. However, at runtime, the Retriever fails to fetch data from these transform-derived tables, quietly returning an empty payload (0 rows) to the Agentforce agent.

The Technical Mitigation Workaround: To bypass this silent runtime bug, a custom DMO named **Consolidate_Web_Data__dlm** was manually created to act as a physical data table. The web activity dataset was loaded directly into this custom DMO, forcing a permanent database index. Creating the Search Index and Retriever over this physical table completely fixed the runtime routing issue, allowing Agentforce to reliably fetch the data.

8. Project Context & Execution Strategy Analysis

Assumptions Made

- **Data Scale Limitations:** Standardized boundaries (such as a max cap of 100 clicks and 10 cart items) are applied to keep calculations accurate and reliable.
- **Static Baseline Records:** Data ingestion is currently processed using static datasets to maintain environment predictability and clear validation loops.
- **Intent Drivers:** Customer Intent dependencies are evaluated based on a combination of the calculated Engagement Score and current CRM Loyalty traits.
- **Input Parameters:** The prompt template accepts only the customer's name string as an input to build next best action recommendations.

Design Decisions

- **Consolidated Data Architecture:** Engineered the Apex controller to pull basic profile fields and clickstream activity within a single, unified ANSI SQL query statement. This design reduces server round-trips and speeds up dashboard loading dramatically.
- **Client-Side Text Sanitization:** Decoupled layout formatting from the backend. The LWC uses a client-side regular expression to intercept text streams and filter out background boilerplate text at runtime, preserving prompt reuse.

Challenges Faced

- **The Data Transform Runtime Defect:** Overcame a critical platform routing bug where retrievers built on top of transformed DMOs successfully indexed but returned empty arrays at runtime. This was bypassed by utilizing a physical custom DMO table instead.
- **Data Representation Alignment:** Faced challenges when structuring test datasets to accurately capture and mimic an authentic user's web engagement footprint across anonymous states.

Trade-offs Considered

- **Physical Table Storage vs. Transforms:** Direct-loading data into a custom DMO uses more storage space than running a dynamic transform pipeline, but it was a necessary trade-off to bypass the platform's runtime retrieval bug and guarantee grounding.
- **Client-Side Cleansing vs. Prompt Restrictions:** Handling the text cleanup via frontend JavaScript requires minor browser processing but keeps the prompt template reusable across other workflows (like SMS bots) that still need the full text string.
- **Calculated Insight Processing Cost:** Scores are calculated at the data stream level which increases platform utilization costs, but this was required since transform-derived DMOs cannot currently feed data to Agentforce.

What You Would Improve with More Time

- **Live Streaming Integration:** Connect the architecture to active live data streams to process large volumes of incoming web events instantly rather than handling batch static file loads.
- **Resource Optimization:** Refine the background infrastructure footprint to maximize server performance and lower processing usage costs under bulk traffic load conditions.

9. Environment Delivery & Access Metadata

	A	B
1	GitHub Repository	<code>https://github.com/devanshkashyap-salesforce/DataCloud_Assignment</code>
2	Salesforce Org URL	<code>https://storm-ecc1703f073cd0.my.salesforce.com</code>
3	Username	<code>devansh.dcassignment@salesforce.com</code>
4	Password	<code>@Salesforce01</code>